

Assignment 4 (Logistic Regression)

Problem Statement

You are given a dataset containing information about users, including their gender, age, and estimated salary. Build a Logistic Regression model to predict whether a user will purchase a product or not. Compute the classification accuracy and Binary Cross-Entropy loss of your result.

Coding Steps

1. **Prepare the dataset:** Read the given training dataset and perform the necessary preprocessing. The dataset contains several input features and a **Purchased** column representing the output.

[Hints: Convert the **Gender** column into a numerical representation. Do not use **User ID** as an input feature, since it is only an identifier.]

2. **Separate the input and output:**

[X = input features]

[y = target variable]

3. **Split the Dataset:** Divide the dataset into training and testing sets. Use 80% of the data for training and 20% of the data for testing. Use a fixed random state so that the result can be reproduced.
4. **Train the Logistic Regression model:** You are allowed to use the Logistic Regression implementation available in a machine learning library, or implement Logistic Regression from scratch.

For Logistic Regression, the probability of one class is calculated using the sigmoid function:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

where, $[x = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots]$

5. **Make predictions:** For each input row in the test dataset, calculate the predicted probability and predicted class. Use a probability threshold of 0.5.
6. **Compute classification accuracy:** Calculate the accuracy of the model using:

$$\text{Accuracy} = \frac{\text{Number of correct outputs}}{\text{Total number of samples}}$$

Print your final accuracy.

7. **Compute Binary Cross-Entropy Loss:** Using the predicted probabilities, calculate the Binary Cross-Entropy (BCE) loss:

$$L_{BCE} = -\frac{1}{n} \sum_{i=1}^n (Y_i \cdot \log \hat{Y}_i + (1 - Y_i) \cdot \log (1 - \hat{Y}_i))$$

Print your final Binary Cross-Entropy loss.